INTRODUCTION

Database Management System (DBMS) and Programming

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Data VS Information

Data

- Raw data is collected from various resources in and out of the organization or even situations occurring in daily life.
- Raw data has no meaning and is not ready to use until it is processed in order to gain the desires outcomes.

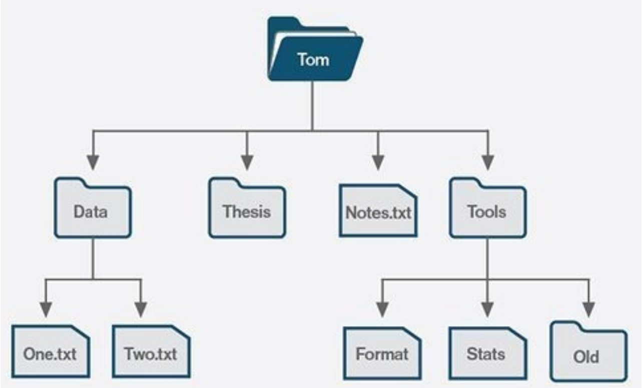
Information

- Information gained through the process in the format that is stored in a systematic way.
- The results can be used to support the decision making.

Data Storage

- Save digital data and information for using later
- Be the back up place to protect the data loss
- Manipulate with the updated data
- There are 2 types
  - File- based
  - Database

File-based



- Also called a file system
- Manage files and directories, and provide basic operations for creating, deleting, renaming, and accessing files.
- Typically built into an operating system.
- Stores and retrieves data from a storage medium like a hard disk or flash drive.

File based vs Database

Human Resource department

Keeps data of
all employees

Finance department

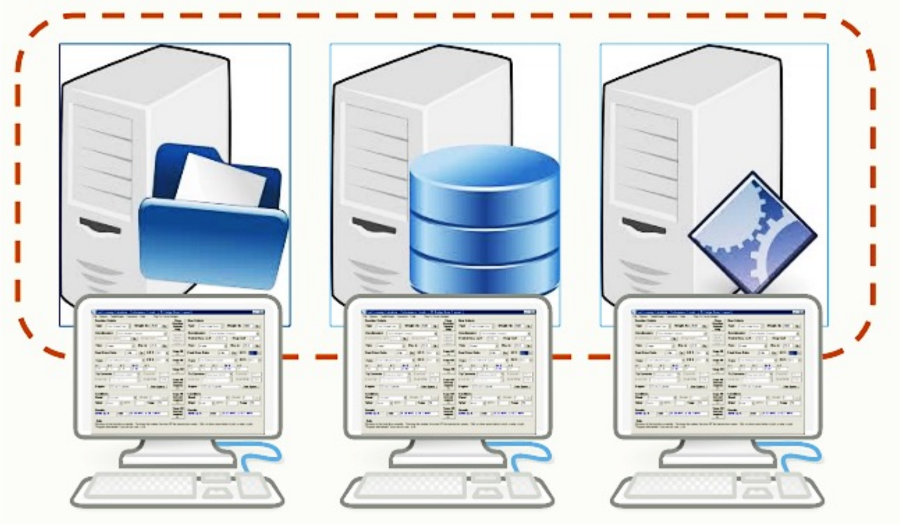
Keeps data of
all employee
salary

Manager

List of managers
and details

Limitations of File Based System

- Separation and isolation data
- Difficult to access the data which is isolated and stored in separate files.



Limitations of File Based System

Duplication of data

- Uncontrolled duplication of data
- Waste of space and it costs time and money to enter the data more than once

player_name	player_avatar	game_number	round_number	gamelink	word	total_point	rank
A	image1.png	1	1	https://skribbl.io/?HnqedHBhPlHH	car	10	3
B	image2.png	1	1	https://skribbl.io/?HnqedHBhPlHH	telephone	90	2
C	image3.png	1	1	https://skribbl.io/?HnqedHBhPlHH	banana	132	1
A	image1.png	1	2	https://skribbl.io/?HnqedHBhPlHH	dog	10	3
B	image2.png	1	2	https://skribbl.io/?HnqedHBhPlHH	fish	90	2
C	image3.png	1	2	https://skribbl.io/?HnqedHBhPlHH	flower	132	1
A	image1.png	2	1	https://skribbl.io/?CsiD6FYlPP3X	camera	9	4
D	image4.png	2	1	https://skribbl.io/?CsiD6FYlPP3X	dog	134	1
B	image2.png	2	1	https://skribbl.io/?CsiD6FYlPP3X	face	50	3
E	image5.png	2	1	https://skribbl.io/?CsiD6FYlPP3X	door	130	2

Limitations of File Based System (cont.)

- **Incompatible format**
  - The structure of files are dependent on the application programming languages.
  - The structure of a file generated by a PYTHON program may be different from the structure of a file generated by a C program.
- **Fixed query**
  - File-based systems are very dependent upon the application developer, who has to write any queries or reports that are required.
- **Data dependence**
  - The data stored in file depends upon the application program through which the file was created

Database

- Database is a collection of related data which stored and accessed electronically.

Characteristic of database system

- **Self-describing nature of database system**
  - Database system contains *metadata* which defines and describes the data and relationships between tables in the database.

emlployee_id	first_name	last_name	nin	department_id
44	Simon	Martinez	HH 45 09 73 D	1
45	Thomas	Goldstein	SA 75 35 42 B	2
46	Eugene	Comelsen	NE 22 63 82	2
47	Andrew	Petculescu	XY 29 87 61 A	1
48	Ruth	Stadick	MA 12 89 36 A	15
49	Barry	Scardelis	AT 20 73 18	2
50	Sidney	Hunter	HW 12 94 21 C	6
51	Jeffrey	Evans	LX 13 26 39 B	6
52	Doris	Berndt	YA 49 88 11 A	3
53	Diane	Eaton	BE 08 74 68 A	1
54	Bonnie	Hall	WW 53 77 68 A	15
55	Taylor	Li	ZE 55 22 80 B	1

Data

Metadata

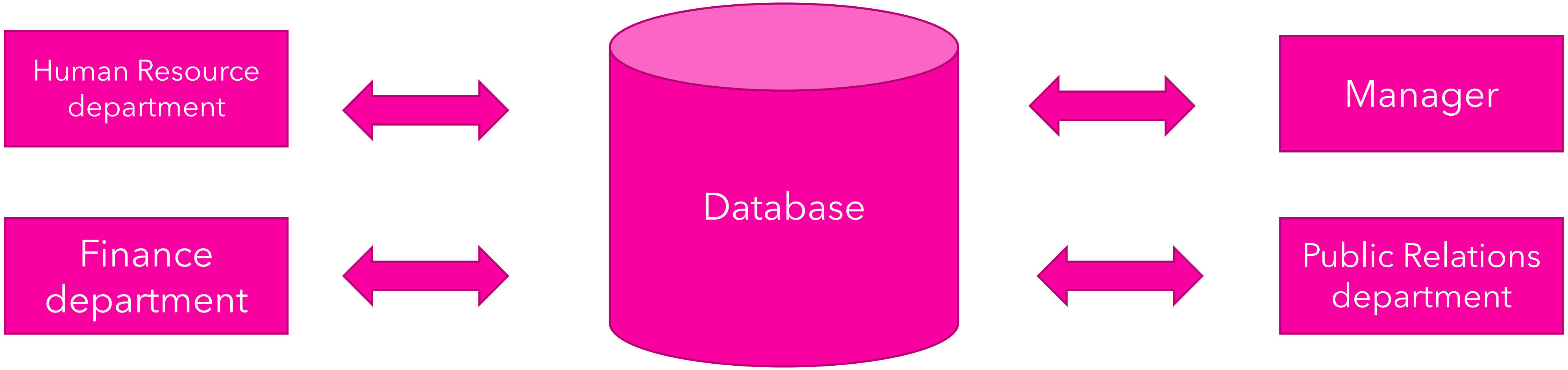
Column	Data Type	Description
emlployee_id	int	Primary key of a table
first_name	nvarchar(50)	Employee first name
last_name	nvarchar(50)	Employee last name
nin	nvarchar(15)	National Identification Number
position	nvarchar(50)	Current postion title, e.g. Secretary
department_id	int	Employee depamtnet. Ref: Departmetns
gender	char(1)	M = Male, F = Female, Null = unknown
employment_start_date	date	Start date of employment in organization.
employment_end_date	date	Employment end date. Null if employee sti

Characteristic of database (cont.)

- **Insulation between programs and data (data independence)**
  - Data structure of database is stored in the system catalogue and not in the programs.
  - So, one change is all that is needed to change the structure of a file.
- **Support multiple views of the data**
  - A *view* is a subset of the database
  - Multiple users in the system might have different views of the system.
- **Sharing of data and supporting multiple user**
  - Many users can access the same database at the same time
  - *Concurrency control strategies* - data accessed are always correct and that data integrity is maintained.

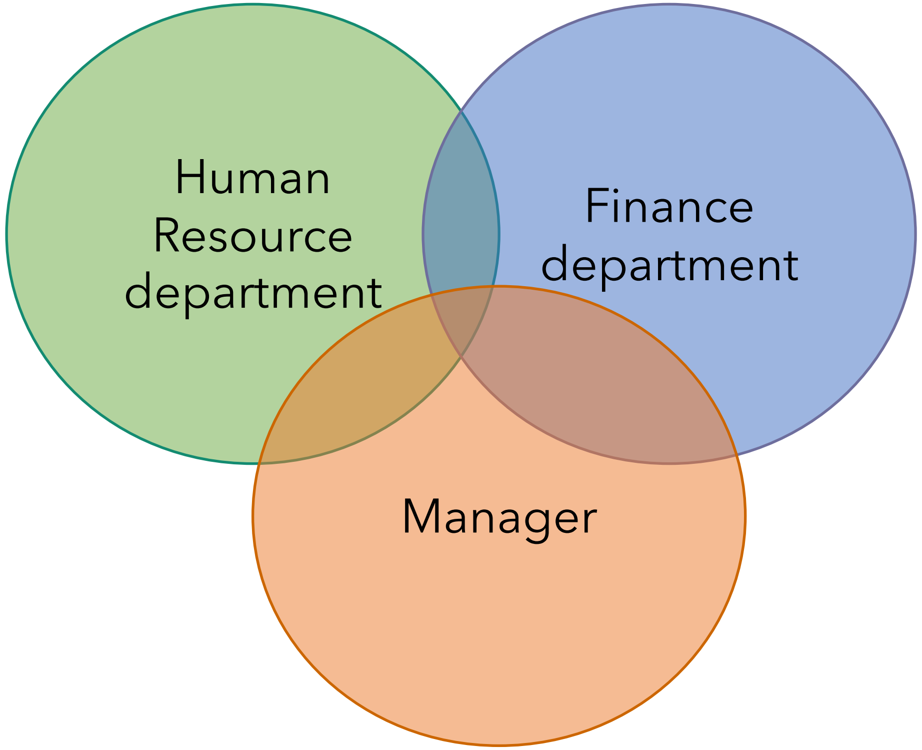
Characteristic of database (cont.)

- Single large repository of data



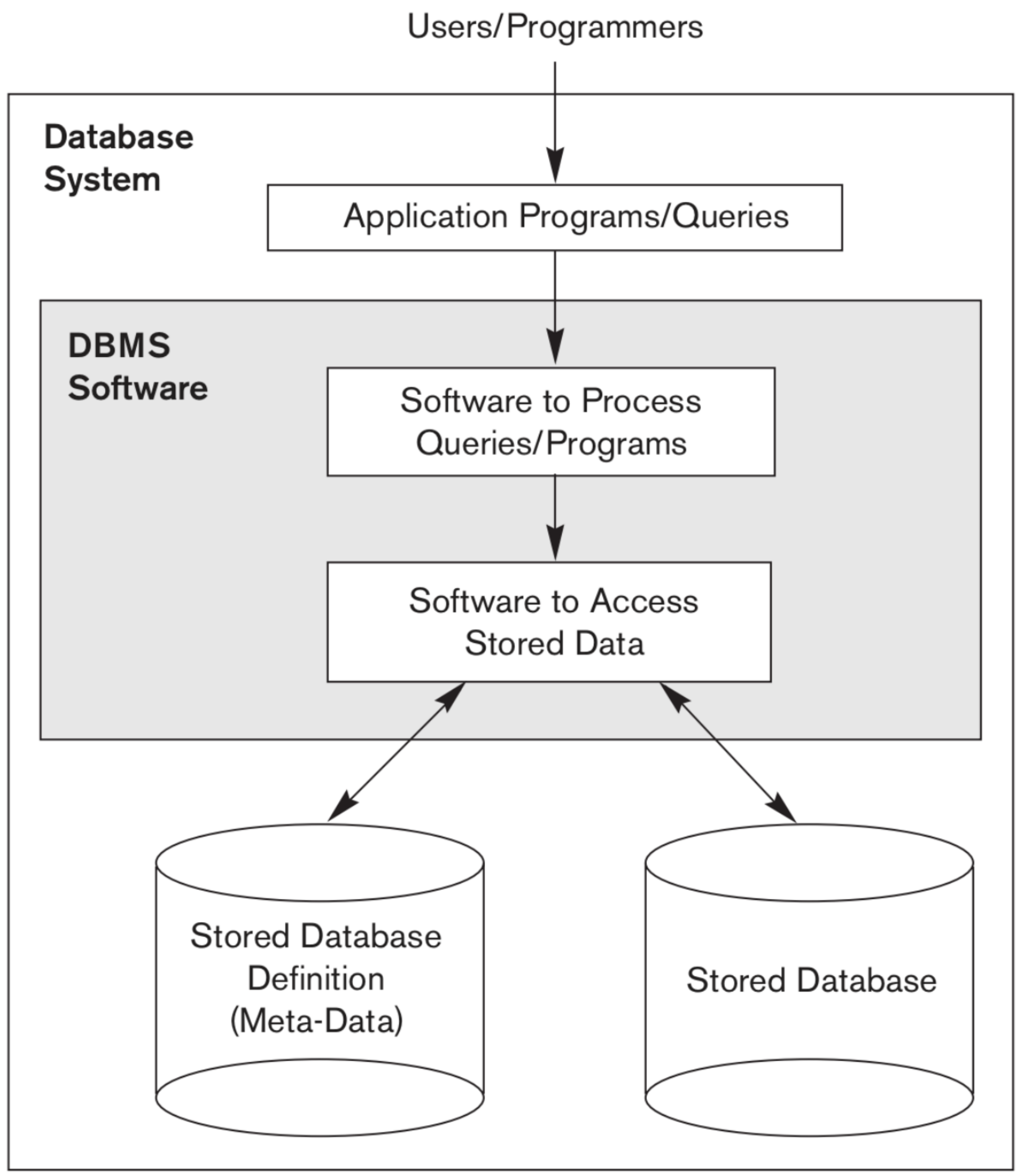
Characteristic of database (cont.)

- Minimize duplication



DATABASE MANAGEMENT SYSTEM (DBMS)

Database Management System (DBMS)

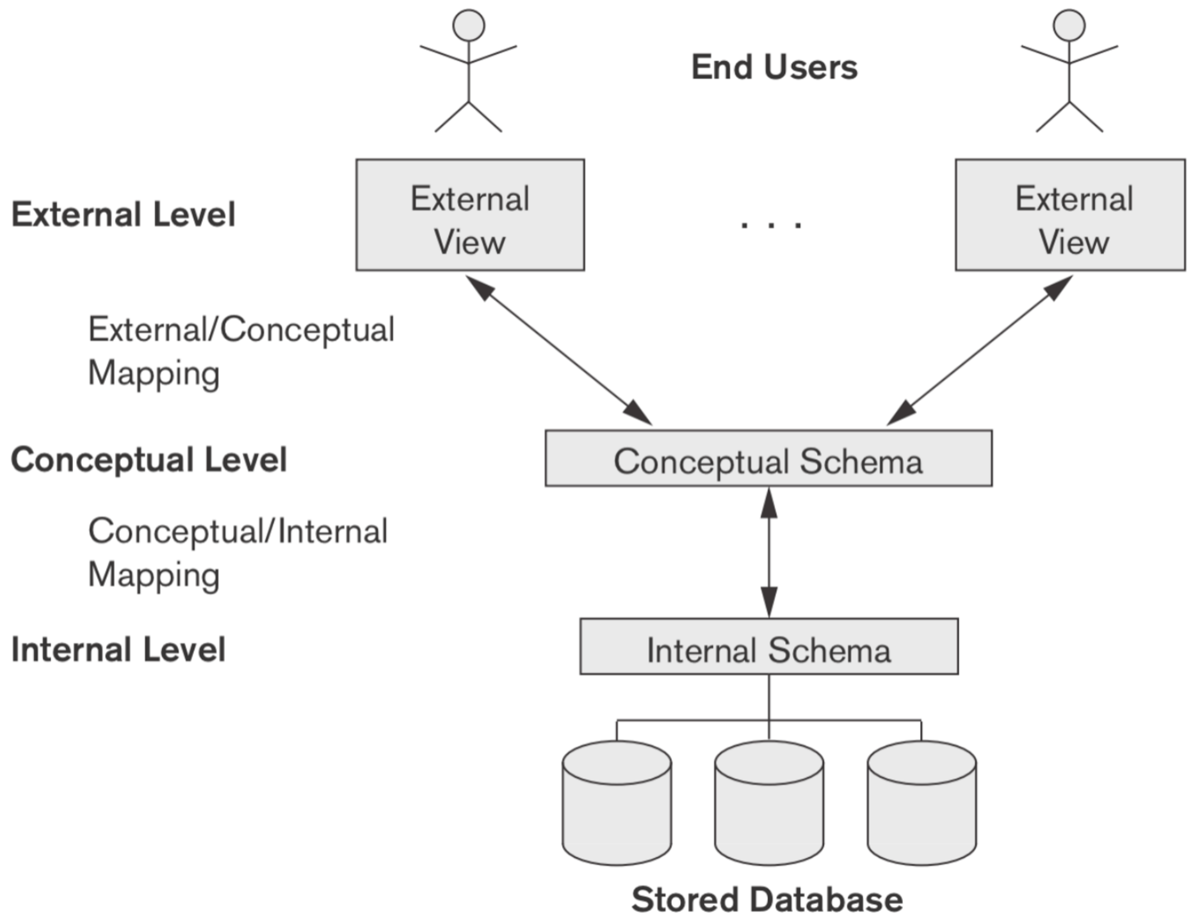


Database Management System (DBMS)

- **Database System** is the DBMS software together with the data itself. Sometimes, the applications are also included.

- **Database Management System (DBMS)**
  - The software that interacts with end users, applications, and the database itself to capture and analyze the data.
  - Enable user to define, create, maintain and control access to the database.

Schema Architecture



Schema Architecture (cont.)

External Level

- The *users' view* of the database.
- Describes the relevance of each user to part of the database.

Conceptual Level

- The *community view* of the database.
- Describes *what* data is stored in the database and relationships among the data.

Internal Level

- The *physical representation* of the database on the computer
- Describe *how* the data is stored in the database.

Database Schemas

- **Database schema** is the description of a database
  - Not change frequently
  - Schema diagram - displays the structure of each record type but not actual instances of records.

Finance

No	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	25000
4	Jennifer	Wallace	43000

Finance

No	Firstname	Lastname	Salary

```
FINANCE{No, Firstname, Lastname, Salary}
```

Database Schemas

HUMAN_RESOURCE

No	Firstname	Lastname	NID

HUMAN_RESOURCE

No	Firstname	Lastname	Address	Gender

HUMAN_RESOURCE

No	Firstname	Lastname	NID	Birthdate	Address	Gender

```
struct HUMAN_RESOURCE {
      int No;
      varchar Firstname [50];
      varchar Lastname [50];
      char NID [13];
      date Birthdate;
      varchar Address [100];
      boolean Gender;
      struct HR *next;     /*pointer to next HR record */
};
index No;
```

Instances

- Collection of information stored at a particular moment.
- Data in instances can be changed using addition, deletion, updating.
- Set of Information stored at a particular time.

Finance

No	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	25000
4	Jennifer	Wallace	43000

```
{1, John, Smith, 30000}
{2, Franklin, Wong, 40000}
```

Database language (DBL)

- **Data Definition Language (DDL)** – A Language that define database *conceptual schema*.
  - CREATE
  - DROP
  - ALTER

- **Data Manipulation Language (DML)** – A Language that manipulate the data held in the database.
  - SELECT
  - INSERT
  - UPDATE
  - DELETE

Functions of DBMS

1. Data storage, retrieval and update

- A DBMS must give the users the ability to store retrieve and update data in the database

2. A user-accessible catalog

- A DBMS must give a data catalog in which descriptions of data items are stored and which is accessible to users.
    - Name, types and size of data items
    - Name of relationships
    - Integrity constrains on the data
    - Name of authorized users who have access to the data

Functions of DBMS

3. Transaction support

- DBMS must give a mechanism which will ensure that all the updates corresponding to a given transaction and made or that none of them is made.

Simple Transaction Example

1. Read your account balance
2. Deduct the amount from your balance
3. Write the remaining balance to your account
4. Read your friend's account balance
5. Add the amount to his account balance
6. Write the new updated balance to his account

```
1. Read(balA);
2. balA = balA - 10000;
3. write(balA);
4. read(balB);
5. balB = balB + 10000;
6. write(balB);
```

Functions of DBMS

4. Concurrency control services

- A DBMS must furnish a mechanism to ensure that the database is updated correctly when multiple users are updating the database concurrently

Time	T1	T2	bal_x
t1		read(bal_x)	100
t2	read(bal_x)	$bal_x = bal_x + 100$	100
t3	$bal_x = bal_x - 10$	write(bal_x)	200
t4	write(bal_x)		90
t5			90

Functions of DBMS

4. Concurrency control services (cont.)

Time	T1	T2	bal_x
t1		write_lock(bal_x)	100
t2	write_lock(bal_x)	read(bal_x)	100
t3	**WAIT**	$bal_x = bal_x + 100$	100
t4	**WAIT**	write(bal_x)	200
t5	**WAIT**	commit/unlock(bal_x)	200
t6	read(bal_x)		200
t7	$bal_x = bal_x - 10$		190
t8	write(bal_x)		190
t9	commit/unlock(bal_x)		190

Functions of DBMS

5. Recovery services

- A DBMS must furnish a mechanism for recovering the database if the device is damaged in any way.
  - System crash
  - Media failure
  - Hardware/software error

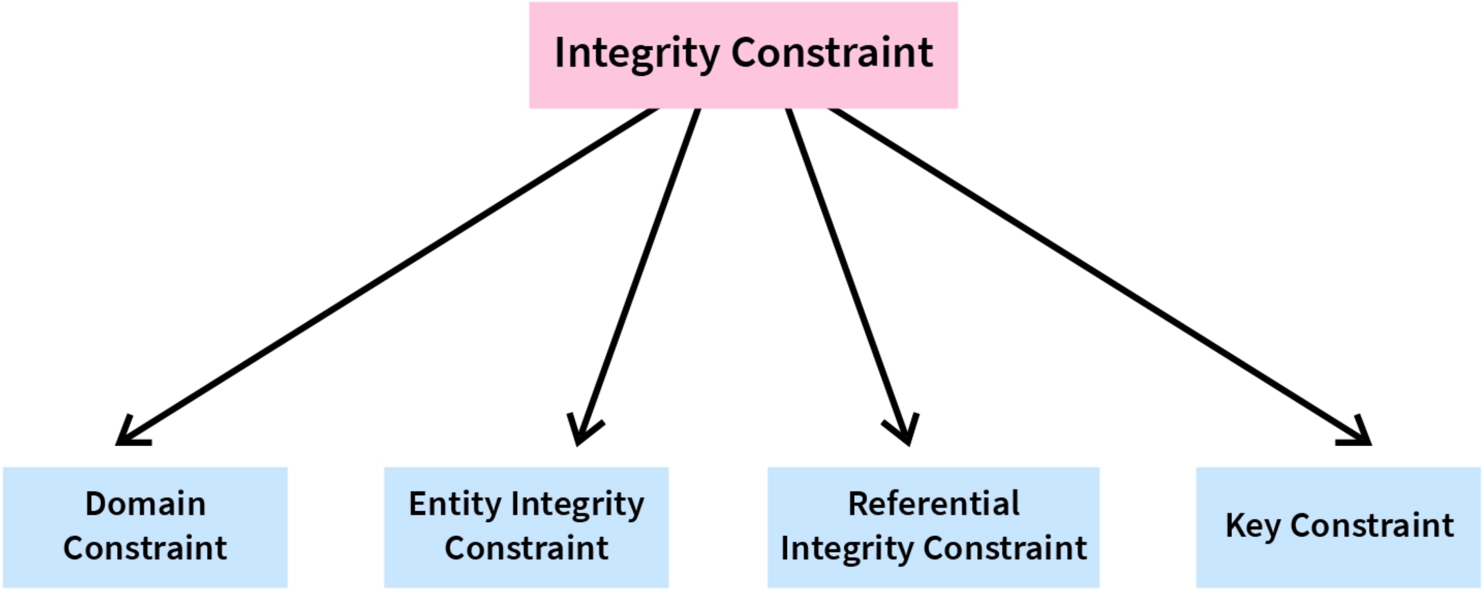
6. Authorization services

- A DBMS must give a mechanism to ensure that only authorized users can access the database

Functions of DBMS

7. Integrity services

- A DBMS must ensure that both the data in the database and changes to the data follow certain rules.
- **The correctness, consistency and completeness of data**



Functions of DBMS

7. Integrity services (cont.)

- **Domain Integrity Constraint** - contains a certain set of rules or conditions to restrict the kind of attributes or values a column can hold in the database table

No	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	A
4	Jennifer	Wallace	43000

Functions of DBMS

7. Integrity services (cont.)

- **Entity Integrity Constraint** - used to ensure that the primary key cannot be null

No	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
	Jennifer	Wallace	43000

Functions of DBMS

7. Integrity services (cont.)

- **Referential Integrity Constraint** - ensures that there must always exist a valid relationship between two relational database tables.

Employee

No	Firstname	Lastname	Salary	Dept_ID
1	John	Smith	30000	1
2	Franklin	Wong	40000	1
3	Alicia	Zelaya	29000	4
4	Jennifer	Wallace	43000	3

Department

Dept_ID	Dept_name
1	Sale
2	HR
3	Research

Functions of DBMS

7. Integrity services (cont.)

- **Key Constraint** - There could be multiple keys in a single entity set, but out of these multiple keys, only one key will be the ***primary key***

No	Firstname	Lastname	Salary
1	John	Smith	30000
2	Franklin	Wong	40000
3	Alicia	Zelaya	29000
2	Jennifer	Wallace	43000

Functions of DBMS

8. Data Independence services

- Each user should be able *to* *access the same data* *but* have *a different customized view* of the data.
- Each user should be able to *change the way he or she views* the data, and this change should *not affect other users*.
- Users should not have to deal directly with physical database storage details, such as indexing or hashing
- The Database Administrator (DBA) should be able to change the database structures without affecting the users' views.
- The internal structure of the database should be unaffected by change structure of storage, such as the changeover to a new storage device.
- The DBA should be able to change the conceptual structure of the database without affecting all users.

Types of DBMS

- **Relational Database**
  - Uses a concept of organizing data into tables, columns and rows.
    - E.g., MySQL, Oracle, Microsoft SQL Server, IBM DB2, MariaDB

- **No-SQL Database (Non-relational Database)**
  - Not organizing data into columns and rows
  - 4 Types of No-SQL Databases
    1. Document Databases - Stores data in JSON, BSON , or XML documents
    2. Key-Value Stores - Stored as a key value pair consisting of an attribute name and a value.
    3. Object-Oriented Databases
    4. Graph Databases
  - E.g., MongoDB, Firebase, Radis